Clinical Study Summary Report (CSSR) Document Status: Draft | **Version:** 1.0 | **Date:** 03-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A Phase I, Open-label, Multi-center Study of Radiation Dosimetry, Safety, and Tolerability of Extended Lutetium (177Lu) Vipivotide Tetraxetan Treatment in Chemo-naïve Adults With Metastatic Castration-resistant Prostate Cancer **Short Title/Acronym:** RADIODOSE Study **Protocol Number:** CAAA617A12101 **NCT Number:** NCT06531499 **Posting ID:** 40732562431363642 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Lutetium (177Lu) vipivotide tetraxetan (also referred to as [177Lu]Lu-PSMA-617 or AAA617) **Phase of Development:** Phase I **Trial Status:** Recruiting **Medical Monitor:** Novartis Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess and evaluate the radiation dosimetry, safety, and tolerability following the administration of up to 12 extended cycles of (177Lu) vipivotide tetraxetan in taxane-naïve adult participants with PSMA-positive metastatic castration-resistant prostate cancer (mCRPC). **Secondary Objectives:** To evaluate radiographic progression, treatment toxicity, pharmacokinetics, changes in PSMA expression levels (re-assessed after Cycle 6), and overall survival/ radiographic progression-free survival (rPFS).

2.2 Background & Rationale To evaluate whether an extended regimen (up to 12 cycles) of targeted radioactive therapy can safely treat advanced prostate cancer that has spread and is no longer responding to androgen receptor pathway inhibitor (ARPI) therapy. Current paradigms utilize fewer cycles; this study investigates extending the therapeutic window in a chemo-naïve population to maximize tumor cell death while carefully monitoring dosimetry and renal/systemic toxicity limits.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (All participants receive the investigational product) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target N: Approximately 106 participants **Number of Sites:** Multi-center (Global sites including US, Germany, Spain, Switzerland, Netherlands, and UK) **Estimated Study Start:** 11-Nov-2024 **Estimated Primary Completion:** 24-Nov-2028

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male participants ≥ 18 years of age with an ECOG performance status ≤ 1. 2. Histological confirmation of adenocarcinoma of the prostate with documented progressive mCRPC, and confirmed PSMA-positive per 68Ga-PSMA PET/CT scans at baseline. 3. Participants must have ≥ 1 metastatic lesion by conventional imaging that is present on screening/baseline CT, MRI, or bone scan. 4. Castrate level of serum/plasma testosterone (< 50 ng/dL or < 1.7 nmol/L) via pharmaceutical or surgical methods, who have progressed only once on prior second-generation ARPIs. 5. Renal: eGFR ≥ 60 mL/min/1.73m2 using the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation. 6. Participants must have recovered to $\le$ Grade 2 from all clinically significant toxicities related to prior therapies except alopecia.

4.2 Exclusion Criteria 1. Previous treatment with any of the following within 6 months of study enrollment: Strontium 89, Samarium-153, Rhenium-186, Rhenium-188, Radium-223, hemi-body irradiation. 2. Any previous radioligand therapy, or prior treatment with cytotoxic chemotherapy for mCRPC/mHSPC (e.g., taxanes, platinum, estramustine, vincristine, methotrexate). Taxane exposure (maximum 6 cycles) in the neoadjuvant/adjuvant setting is allowed only if > 12 months have elapsed. 3. Concurrent therapies: cytotoxic chemotherapy, immunotherapy, alternative radioligand therapy, PARP inhibitors, biological, or investigational therapies. 4. Concurrent serious acute or chronic nephropathy and/or moderate to severe renal impairment as determined by the principal investigator. 5. History of myocardial infarction (MI), angina pectoris, or coronary artery bypass graft (CABG) within 6 months prior to ICF signature and/or clinically active significant cardiac disease. 6. Diagnosed with other active malignancies that are expected to alter life expectancy or may interfere with disease assessment. 7. Concurrent urinary outflow obstruction or unmanageable urinary incontinence. 8. History of somatic or psychiatric disease/condition that may interfere with the aims and assessments of the study. 9. Sexually active males unwilling to use a condom during intercourse while taking study treatment and for 14 weeks after stopping study treatment.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: 7.4 GBq (200 mCi) ($\pm 10\%$) administered every 6 weeks for up to 12 cycles. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Up to 74 weeks of treatment duration (maximum 12 cycles).

5.2 Concomitant & Prohibited Therapy Permitted: Gonadotropin-releasing hormone (GnRH) analogues/antagonists to maintain castrate testosterone levels (e.g., ganirelix [Trade Names: Antagon, Fyremadel; ATC Code: H01CC01]). **Prohibited:** Concurrent cytotoxic chemotherapy, immunotherapy, alternative radioligand therapy, PARP inhibitors, or other investigational therapies.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, baseline 68Ga-PSMA PET/CT scan, renal function testing (eGFR)
Baseline	Day 0	First Dose of 177Lu-PSMA-617 (7.4 GBq)
Interim	Weeks 6, 12, 18, 24, 30, 36	Cycles 2 through 6. **Post-Cycle 6:** Additional PSMA-PET scan to re-assess PSMA expression and eligibility for further treatment
Extended Tx	Weeks 42 to 74	Cycles 7 through 12 (every 6 weeks, if eligible)
End of Study	End of Treatment (EOT) + 42 Days	EOT PSMA-PET scan, 42-day safety visit, survival and rPFS follow-up

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **Time activity curves (TACs) and absorbed radiation dose of AAA617 in organs:** Generation of TACs from tissue radioactivity to calculate time integrated activity coefficients and absorbed dose. * **Incidence and severity of Adverse Events (AEs) and Serious Adverse Events (SAEs):** Evaluated through relevant clinical and laboratory safety parameters. * **Percentage of participants with AAA617 dose reductions, interruptions and discontinuations:** To assess tolerability based on frequency of these events due to toxicity.

7.2 Secondary & Exploratory Endpoints * Time activity curves (TACs) and absorbed radiation dose of AAA617 in tumors. * Pharmacokinetic (PK) concentration of AAA617 in blood over time. * Pharmacokinetic (PK) parameter Cmax of AAA617 from blood radioactivity data. * Radiographic Progression-Free Survival (rPFS) and Overall Survival (OS). * Re-assessment of PSMA expression levels after 6 cycles to determine extended treatment utility.

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard continuous AE collection during all visits and extending for at least 42 days following the end of treatment (EOT) visit. Participants will continue to be followed for safety every 12 weeks during the long-term follow-up for selected AEs. **Serious Adverse Events (SAEs):** Reported within 24 hours (standard Phase I oncology protocol). **Data Monitoring Committee (DMC):** Internal/external safety committee reviewing radiation dosimetry profiles to ensure organ toxicity thresholds are not breached during extended cycling.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants receiving ≥ 1 dose), Dosimetry Set (participants with sufficient imaging data for exposure calculations). **Sample Size Justification:** $N=106$ provides sufficient power for a Phase I dosimetry and safety evaluation of an extended radioactive cycling regimen. **Handling of Missing Data:** Standard survival analysis methodologies (e.g., right-censoring for rPFS at the date of the last evaluable tumor assessment).

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard oncology Phase I monitoring with specialized radiopharmaceutical oversight at multi-center institutions. **Audit Trail:** Data collection via eCRF including secure transfer of local/central baseline and post-treatment PSMA-PET imaging.

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11. Study Identifiers & Governance

Primary ID: CAAA617A12101 **Registry Number**: NCT06531499 **Sponsor/ Lead Organization**: Novartis Pharmaceuticals **Study Title:** RADIODOSE **Official Regulatory Title**: A Phase I, Open-label, Multi-center Study of Radiation Dosimetry, Safety, and Tolerability of Extended Lutetium (177Lu) Vipivotide Tetraxetan Treatment in Chemo-naïve Adults With Metastatic Castration-resistant Prostate Cancer

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12. Clinical Framework (Radioligand Therapy Specific)

Primary Condition: Metastatic Castration-Resistant Prostate Cancer (mCRPC) **Disease Classifications**: SNOMED ID: 789498007 | MeSH Code: D000085434 **Diagnosis Threshold**: PSMA-positive (via 68Ga-PSMA PET/CT) and castrate testosterone levels (< 50 ng/dL) **Medical Methodology:** Targeted Radioligand Therapy **Optimization Target:** Radiation dose delivery to tumor cells while minimizing off-target toxicity over an extended 12-cycle regimen

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13. Study Procedures & Methodology

Management Pathway: Up to 12 cycles of 7.4 GBq AAA617, with a critical re-evaluation gateway after Cycle 6. **Education Protocol (HCP):** Radiation safety handling, dispensing protocols, and dosimetry capture training. **Education Protocol (Patient):**

Radiation safety precautions post-infusion, mandatory contraceptive protocols (condom use required during treatment and for 14 weeks after stopping treatment). **Quality Audit Mechanism:** Centralized imaging reviews and stringent dosimetry tracking.

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14. Observation & Follow-Up Schedule

Total Duration: Up to 74 weeks of active treatment + long-term post-treatment follow-up (every 12 weeks for selected safety parameters). **Onsite Visit Cadence:** Every 6 weeks (Q6W) during the treatment phase. **Remote Monitoring Frequency:** Intermittent survival and safety tracking post-EOT. **Checklist Review Interval:** Prior to every 6-week infusion to verify toxicity limits and dosing safety.

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]